HAM-CD: Hybrid Attention Mamba for Remote Sensing Change Detection

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***Abstract*—The Mamba-based change detection (CD) methods have achieved remarkable success owing to their ability to model long-range pixel dependencies with linear time complexity effectively. While effective in modeling long-range pixel relationships, these methods exhibit limitations in comprehensively learning complex image spatial structures and fully capturing local pixel dependencies critical for effective spatial–temporal representation learning and accurate CD. To overcome this drawback, we propose a hybrid attention Mamba CD (HAM-CD) method that achieves efficient global context modeling while simultaneously maintaining fine-grained local detail representation. Our HAM-CD employs the visual Mamba architecture as its encoder, effectively capturing both global and local spatial–temporal features from the input image. To enhance both the global and local representation ability of the visual Mamba, we design a decoder composed of several hybrid attention Mamba (HAM) blocks, which integrates the transposed attention block (TAB) into the textual-aware Mamba (TAM) block. The HAM effectively captures spatial–temporal feature interactions by utilizing the TAM to strengthen local pixel dependencies and the TAB to enhance global dependencies. Additionally, to mitigate error propagation across stages and enhance more accurate CD, we introduce an interactive channel–spatial fusion (ICSF) block that improves cross-stage feature interaction by jointly modeling spatial- and channel-wise dependencies. Comprehensive experiments on three benchmark datasets demonstrate that our HAM-CD consistently outperforms existing methods, achieving state-of-the-art (SOTA) results without requiring sophisticated training schemes or specialized optimization techniques. Our source code and pretrained models are publicly available at https://github.com/guanguanboy/HAM-CD**

***Index Terms*—Binary change detection (CD), hybrid attention Mamba (HAM), remote sensing, state-space model.**

I. Introduction

CHANGE detection (CD) has been a highly regarded and actively explored area within the remote sensing community ever since the emergence of remote sensing technology. Its objective is to identify alterations in objects present on the Earth's surface by analyzing multitemporal images that are captured at various time intervals. Currently, CD techniques hold a pivotal position in numerous domains, such as the analysis of land-cover changes, investigations into urban expansion, disaster management and response, the updating of geographic information systems, and ecological surveillance [1], [2].

Regarding CD, traditional pixel-based CD methods [3], [4], [5] often struggle to yield satisfactory detection outcomes. In response to this challenge, researchers have proposed object-based CD methods [6], [7], [8], [9]. These methods rely on objects as the basic units for CD. However, these methods are based on hand-crafted features. Such features are insufficiently representative and robust to effectively handle the intricate environmental variations present in multitemporal images [10], [11]. The advent of deep learning has introduced novel approaches for CD, substantially boosting its efficiency and accuracy. Since fully convolutional neural networks (CNNs) were first introduced into the CD field by Daudt et al. [12], CNN-based approaches have consistently demonstrated superior performance in CD research. The integration of computer vision architectures with domain-specific CD constraints has yielded numerous influential methods [13], [14], [15], [16]. However, the constrained receptive field of convolutional operations fundamentally restricts modeling of global contextual relationships, leading to suboptimal performance when processing heterogeneous multitemporal data with varying spatial and temporal characteristics [17]. The Vision Transformer [18] overcomes CNN's fundamental constraints by inherently modeling global spatial relationships through multi-head attention. Thus, contemporary CD methods increasingly employ Transformer-based encoders for discriminative feature extraction [13], [14], [15], [16], followed by spatial–temporal relationship modeling in decoder networks through cross-temporal feature interaction [17], [19], showing excellent

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performance. Besides, Yang et al. [20] proposed SeLHIB, which pioneers the integration of higher-order information bottleneck theory into self-supervised spiking neural networks (SNNs). As an alternative to Transformers, the Mamba architecture [21] has demonstrated state-of-the-art (SOTA) performance in capturing long-range dependencies with only linear growth in computational requirements. Combined with hardware-aware implementations, Mamba architecture outperforms Transformers in efficiency and effectiveness for many downstream tasks, including CD. ChangeMamba [22] stands as the most prominent Mamba-based approach for CD. However, ChangeMamba lacks robust modeling of fine-grained local spatial relationships across multitemporal images. In addition, existing multistage CD methods typically adopt a cascaded pipeline, where each stage incrementally refines the features based on the output of the preceding stage. In such designs, information loss inevitably occurs at each stage. As a result, this lossy feature-level data flow leads to the propagation and accumulation of errors across stages, particularly in regions with subtle or ambiguous changes. To overcome these limitations, we propose a hybrid attention Mamba CD (HAM-CD) method to effectively extract robust global–local spatial–temporal representations from multitemporal images. HAM-CD seamlessly integrates the transposed attention mechanism [23] into the spatial-Mamba framework [24]. This integration leverages the global contextual modeling capabilities of transposed attention and the fine-grained local pixel relationship learning power of spatial-Mamba. Specifically, the recently proposed spatial-Mamba adopts a structure-aware state fusion equation to capture spatial structural relationships within images and constructs local interactions explicitly within the state space. Building upon this, HAM-CD enhances the ability to model local pixel dependencies. As a result, the proposed method effectively captures both global and local spatial relationships in multitemporal images and extracts robust spatial–temporal representations, leading to improved CD performance. The synergistic integration of spatial-Mamba and transposed attention within our HAM-CD framework forms a novel paradigm that significantly enhances the effectiveness and robustness of CD. Besides, we introduce an interactive channel–spatial fusion (ICSF) block, which enables cross-stage feature interaction by jointly modeling spatial- and channel-wise dependencies. This design effectively mitigates error propagation across stages while enhancing the aggregation of complementary multilevel features. The resulting robust spatial–temporal representations preserve fine-grained textual details and maintain structural consistency, ultimately yielding higher-fidelity CD maps. Experimental results on the three CD benchmark datasets validate the superior performance of our HAM-CD. Moreover, our HAM-CD exhibits superior computational efficiency, achieving an optimal tradeoff between model performance and operational efficiency, as illustrated in Fig. 1. The principal contributions of this work are as follows.

1) We propose a novel HAM-CD method that integrates the Mamba architecture and transposed attention to enhance the learning of robust spatial–temporal representation and enable effective and efficient remote sensing CD.
2) We introduce the ICSF block to effectively integrate pretime and posttime feature representations and reduce structure information loss by jointly modeling their differences in both channel- and spatial-wise dimensions.
3) Extensive experiments on three standard benchmark datasets and ablation studies show that HAM-CD outperforms existing CD methods, demonstrating its superior capability in learning robust spatial–temporal representations.

This article is structured as follows. Section II presents a comprehensive literature review of relevant studies. The methodological framework of our proposed HAM-CD model for binary CD is elaborated in Section III. Section IV analyzes and discusses the experimental findings. Finally, Section VI concludes the article with key contributions.

II. Related Works

For remote sensing image processing, CNNs and Transformer-based architectures have achieved remarkable progress in modeling spatial–temporal dependencies. Beyond these mainstream paradigms, SNNs [25], [26], [27] have recently emerged as a promising alternative due to their event-driven, energy-efficient computing paradigm and inherent capability to model fine-grained spatial–temporal dynamics. Most recently, Mamba-based architectures have been increasingly adopted in remote sensing image analysis, particularly demonstrating strong potential in the field of CD. Accordingly, this section presents a thorough survey of SOTA deep learning paradigms for binary CD, with a particular focus on CNN-, Transformer-, and Mamba-based approaches.

A. CNN-Based Approaches

The rapid advancement of CNNs and their remarkable ability to automatically extract hierarchical features have led to their widespread application in CD tasks. Initially, the patch-based framework [28], [29] was dominant. Daudt et al. [12] made a pioneering effort by introducing fully CNNs to the CD tasks. Following this, a range of influential methods were proposed. Chen et al. [30] introduced a framework that integrates a long short-term memory (LSTM) mechanism with CNNs. Additionally, Fang et al. [31] proposed a Siamese network characterized by dense connections, which enhanced the exchange of shallow information in BCD tasks. Zhao et al. [32] examined the impact of different fusion strategies on BCD performance, proposing a dual encoder–decoder architecture that aimed to improve detection efficacy. In addition to tackling the issue of limited sample availability, transfer learning techniques were introduced. Guo et al. [33] developed an attention-based network for semi-supervised building update detection. The development of the CNN-based CD methods has advanced considerably over time. Nonetheless, despite their remarkable capacity for extracting local pixel relationships, CNN architectures are inherently limited in capturing long-range pixel relationships [18]. This constraint becomes particularly problematic in CD tasks, where the changed regions are

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often sparse and dispersed, in contrast to the more uniform background areas. Precise CD fundamentally requires the network to effectively model long-range relationships that extend beyond local regions [13], [17].

B. Transformer-Based Approaches

Recent advancements have demonstrated the impressive potential of Transformers in the domain of computer vision [18], [34]. Their superior ability to model long-range dependencies has enabled them to overcome the limitations faced by CNNs. Consequently, numerous studies have turned to Transformer-based architectures to address CD tasks. Bandara and Patel [13] proposed a fully Transformer-based Siamese network, employing an encoder paired with a multilayer perceptron decoder, eliminating the need for CNN-based feature extraction. Similarly, Zhang et al. [15] presented a purely Transformer-driven model with a Siamese U-shaped structure, enabling the full recovery of detailed change information. Li et al. [19] introduced a hybrid architecture that combines the global pixel dependencies modeling strength of Transformers with the U-Net architecture. This architecture eliminates redundant U-Net features while augmenting discriminative characteristics via a differential refinement component, yielding superior detection precision. While Transformer-based methods have notably boosted the accuracy of CD tasks through their robust nonlocal modeling capabilities, they are accompanied by significant computational overhead. The self-attention (SA) mechanism exhibits quadratic complexity growth relative to pixel count, creating significant bottlenecks for high-resolution image CD. Recently, the emerging Mamba architecture [21] has demonstrated comparable nonlocal modeling performance to Transformers, while maintaining a linear growth rate in computational overhead. This breakthrough facilitates the potential for both efficient and accurate CD, addressing the computational challenges typically associated with Transformer-based approaches. In this article, we introduce a Mamba-based approach to effectively solve the binary CD task.

C. Mamba-Based Approaches

An alternative to the Transformer in analyzing lengthy sequential data is the Mamba model [21], which provides the advantage of linear computational scaling. Currently, the Mamba architecture has been adapted for computer vision tasks [35], [36], demonstrating competitive performances. Inspired by these advancements, Mamba has been extended to the remote sensing image processing [37], [38] and the CD domain [22], achieving remarkable performance. However, the original Vision Mamba [21] is limited in model effectiveness and efficiency because the directional scanning mechanism disrupts the intrinsic spatial information within images. Consequently, these methods demonstrate limited effectiveness in capturing comprehensive and dynamic spatial computations. In this article, we introduce a hybrid attention Mamba (HAM) architecture for CD tasks, designed to effectively model both global and spatial dependencies among neighboring features.

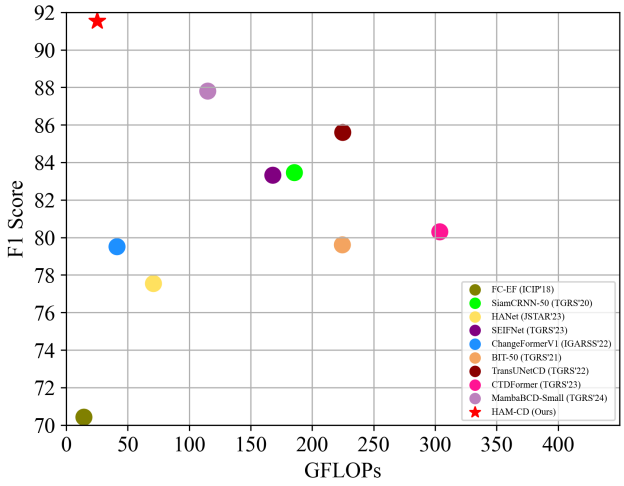


Fig. 1. Tradeoff between computational efficiency and CD performance. As shown, our HAM-CM strikes an optimal balance between performance and inference efficiency.

III. Methodology

A. Overall Structure

Based on the shared characteristics and inherent properties of binary CD tasks, we propose HAM-CD, a novel Mamba-based architecture enhanced with integrated attention mechanisms. The overall architecture of HAM-CD consists of an encoder and a decoder, as depicted in Fig. 2. The encoder adopts a weight-sharing Siamese network, constructed using the visual state-space model (VMamba) [35]. VMamba leverages the advantages of the Mamba framework along with an efficient 2-D cross-scan mechanism to extract robust multiscale features, which are beneficial for the CD task. The decoder is also a Mamba-based network, which is designed to learning the spatial relationships from the extracted features derived from the encoder and detect the changes. The difference in the decoder lies in the fact that it is composed of newly designed HAM blocks, which can better capture the global and local relationships among pixels in the image. The implementation of the encoder and the decoder will be elaborated in Sections III-B and III-C. Given two input images $\mathbf{I}_{t_1}, \mathbf{I}_{t_2} \in \mathbb{R}^{H\times W\times 3}$ captured at different times t_1, t_2, the proposed method first utilizes the Siamese encoder $\mathcal{F}_\theta^{enc}$ to progressively capture hierarchical feature representations across multiple scales. This process can be formulated as

$$\left\{\mathbf{F}_{t_1}^j\right\}_{j=1}^4 = \mathcal{F}_\theta^{enc}\left(\mathbf{I}_{t_1}\right), \quad \left\{\mathbf{F}_{t_2}^j\right\}_{j=1}^4 = \mathcal{F}_\theta^{enc}\left(\mathbf{I}_{t_2}\right). \tag{1}$$

The extracted multilevel features $\{\mathbf{F}_{t_1}^j\}_{j=1}^4, \{\mathbf{F}_{t_2}^j\}_{j=1}^4$ are then fed into the HAM change decoder $\mathcal{F}_\theta^{dec}$ to fully exploit spatial–temporal dependencies using the HAM blocks. This enables the model to refine the change map predictions progressively, formulated as

$$\mathbf{A} = \mathcal{F}_\theta^{dec}\left(\left\{\mathbf{F}_j^{\mathrm{T}_1}\right\}_{j=1}^4, \left\{\mathbf{F}_j^{\mathrm{T}_2}\right\}_{j=1}^4\right). \tag{2}$$

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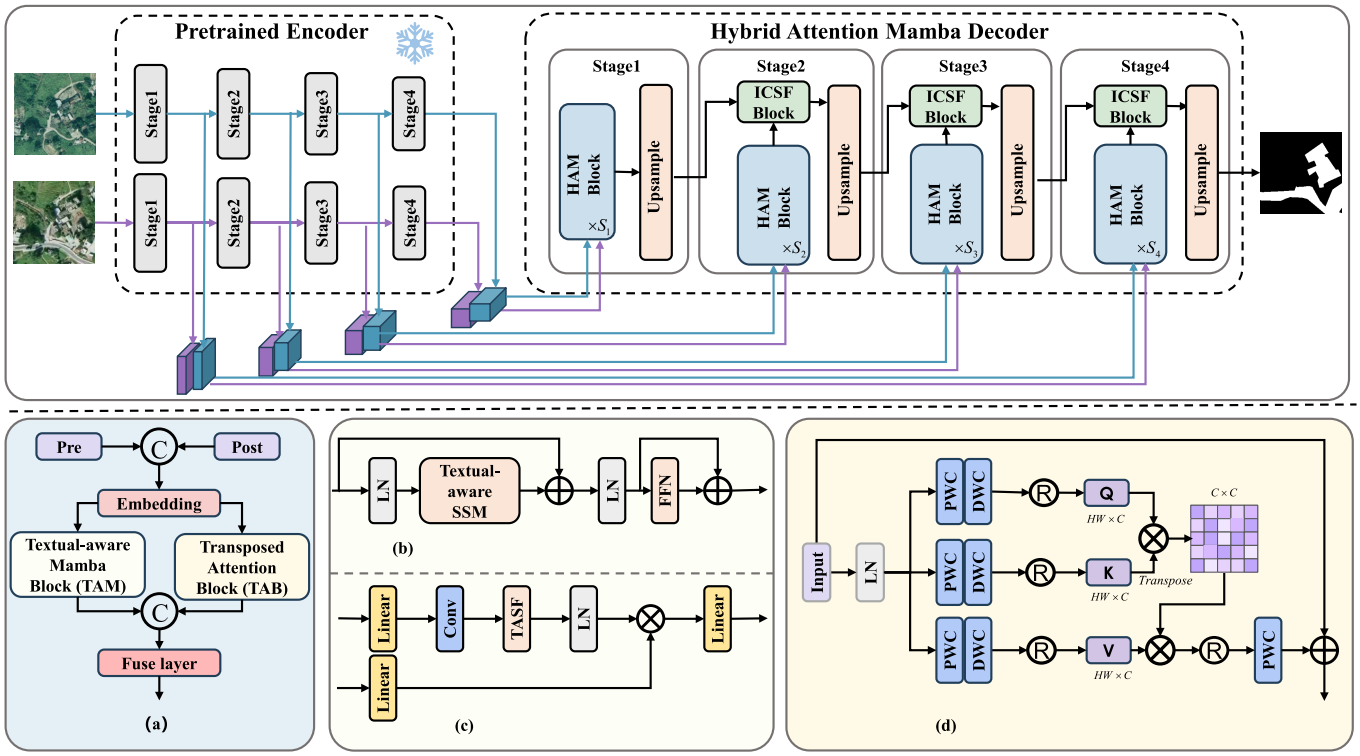


Fig. 2. Overall network architecture of our hybrid attention Mamba (HAM-CD) method. It comprises a weight-sharing Siamese encoder (VMamba-based) and a Mamba-based decoder with HAM blocks. The encoder extracts multitemporal features, while the decoder enhances spatial relationship learning through global–local feature interaction. The ICSF block adaptively fuses cross-stage features to enhance robust spatial–temporal feature representation. (a) HAM. (b) TAM block. (c) Textual-aware SSM. (d) TAB.

Finally, the change map is generated through an argmax operation

$$\hat{\mathbf{Y}}^{bcd} = \text{argmax}_c \mathbf{A}. \tag{3}$$

This process ensures an accurate and reliable binary CD by leveraging the strengths of the HAM in modeling spatial–temporal correlations.

B. Pretrained Encoder

The Mamba architecture inherently adopts a causal processing paradigm, whereby input data is sequentially scanned and processed. This characteristic confines the model's ability to perceive contextual information solely within the traversed data segments. While such a mechanism is well-suited for sequential modalities like natural language, it poses inherent limitations when applied to noncausal data types such as images [35]. A simple strategy of reshaping image data along a single spatial axis and feeding it directly into the S6 model leads to suboptimal contextual modeling, as critical spatial dependencies are inadequately captured. To overcome this challenge, VMamba introduces a 2-D cross-scan scheme that significantly enhances the integration of spatial information. This mechanism reorders image tokens across multiple directional paths before they enter the S6 model. The features derived from these varied traversals are subsequently fused, thereby enabling each pixel to aggregate spatial cues from diverse orientations. Notably, this cross-scan augmentation preserves the linear computational complexity of the original VMamba design, maintaining a complexity of $O(N)$, in contrast to the quadratic cost incurred by Transformer-based SA.

Building upon the VMamba framework, the encoder architecture employed in our method is depicted in Fig. 2. The encoder comprises four hierarchical stages. Each stage begins with a downsampling operation, followed by comprehensive spatial context modeling via a series of visual state-space blocks, and concludes with the generation of intermediate features for both pretime and posttime images, denoted as $\mathbf{F}_{i,j}^{\mathrm{T}_1}$ and $\mathbf{F}_{i,j}^{\mathrm{T}_2}$, respectively. The multiscale features extracted across the four stages, $\{\mathbf{F}_{i,j}^{\mathrm{T}_1}\}_{j=1}^{4}$ and $\{\mathbf{F}_{i,j}^{\mathrm{T}_2}\}_{j=1}^{4}$, are subsequently utilized by the change decoder.

C. Hybrid Attention Mamba Decoder

While the Mamba-based encoder effectively extracts robust and representative features, the decoder also comprehensively models global–local pixel relationships in multitemporal images to achieve optimal CD performance. To this end, we design the HAM decoder to fully exploit the global and local pixel relationships and spatial–temporal dependencies. The overall structure of the HAM decoder is illustrated to the upper right of Fig. 2. It is also composed of four sequential stages. The first stage consists of a single HAM block followed by an upsampling layer with a scale factor of 2. Each of the subsequent three stages comprises an HAM block and an ICSF block, followed by an upsampling operation. Specifically, each processing stage initiates with spatial–temporal relationship modeling of multitemporal features through a

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hybrid attention module block. The representations generated by the HAM at the current stage are subsequently combined with the preceding stage's feature maps via the ICSF block. The integrated features are then passed through an upsample operation before being propagated to the subsequent processing stage. In the context of CD, the HAM can extract both long-range global and local pixel dependencies, enabling a comprehensive feature representation that captures subtle variations at different scales. This capability allows HAM to effectively identify both fine-grained local changes and large-scale semantic shifts, significantly improving detection accuracy. Besides, the ICSF block is specifically designed to deeply fuse the output features from the preceding stage with the current stage's representations, thereby enhancing the hierarchical integration of spatial and contextual information. It comprehensively models the spatial–temporal dependencies between the extracted multitemporal features $\{\mathbf{F}_{i,j}^{\mathrm{T}_1}\}_{j=1}^{4}$ and $\{\mathbf{F}_{i,j}^{\mathrm{T}_2}\}_{j=1}^{4}$ across four hierarchical stages, thereby enabling the generation of accurate binary change maps. By employing the aforementioned feature fusion process, our HAM combines global and local dynamic feature aggregation. This integration guarantees that the final feature representation comprehensively encodes both global contextual patterns and localized spatial relationships, effectively capitalizing on the complementary advantages of each branch. In general, the proposed HAM leads to robust spatial–temporal feature representations.

1) Hybrid Attention Mamba (HAM): The HAM block consists of a textual-aware Mamba (TAM) block and a transposed attention block (TAB), as presented in Fig. 2(a). The TAM block augments the original Mamba's [21] capacity for local pixel relationship modeling while preserving its inherent global dependency modeling capabilities. Meanwhile, the TAB operates along the channel dimensions, augmenting the network's capacity for nonlocal relationship modeling through adaptive channel weighting. We adopt a similar structure of spatial-Mamba [24] to build HAM, but enhance the representative ability of the Mamba by introducing a TAB into it. To avoid potential optimization conflicts between the two key components, we adopt a parallel integration strategy with a learnable fusion layer to adaptively balance the HAM block's outputs. Specifically, the outputs of TAB and TAM complement each other. TAB provides globally consistent contextual representations that guide TAM in refining local details, while TAM contributes critical local textural cues to TAB's representations, which improves the discrimination of subtle changes. Through this bidirectional reinforcement of global and local modeling, the network generates features that are both contextually coherent and detail-preserving, ultimately enhancing the accuracy and robustness of CD. Additionally, the fusion layer integrates these features to generate discriminative spatial–temporal representations for CD. By combining multilevel spatial attention with channel-wise adaptive weighting, HAM enhances discriminative feature learning while suppressing irrelevant background noise, making it particularly advantageous for handling complex change scenarios. Overall, for the given input features X_{pre} and post, the whole process of HAM is defined as

$$\begin{aligned}\mathbf{F}_{\mathrm{emb}} &= \mathrm{EMLayer}\left(\mathrm{Concat}\left(X_{\mathrm{pre}}, X_{\mathrm{post}}\right)\right)\\ \mathbf{F}_{\mathrm{tam}} &= \mathrm{TAM}\left(\mathbf{F}_{\mathrm{emb}}\right)\\ \mathbf{F}_{\mathrm{tab}} &= \mathrm{TAB}\left(\mathbf{F}_{\mathrm{emb}}\right)\\ \mathbf{F}_{\mathrm{fuse}} &= \mathrm{Fuse}\left(\mathrm{Concat}\left(\mathbf{F}_{\mathrm{tam}}, \mathbf{F}_{\mathrm{tab}}\right)\right)\end{aligned} \tag{4}$$

where EMLayer denotes the embedding layer used to transform input features into a dense vector representation. TAM and TAB represent the TAM block and the TAB, respectively. Fuse stands for the feature fusion layer, which is implemented by a convolution operation. The HAM module seamlessly fuses the textural-aware Mamba TAM and TAB paradigms into a cohesive architecture, which yields substantial gains in CD effectiveness and robustness, outperforming prior works. We elaborate on the detailed structures of the TAM and the TAB in the following paragraphs.

a) TAM: The TAM block forms the fundamental building unit of our HAM, which consists of a textual-aware SSM and a feed-forward network (FFN) with skip connections, each preceded by layer normalization, as illustrated in Fig. 2(b). TAM is specifically designed to model spatial dependencies among neighboring features within the latent state space. Building upon the Mamba block design [21], the textual-aware SSM, illustrated in Fig. 2(c), is implemented by substituting the original 1-D causal convolution with a 3×3 depthwise convolution and replacing the original S6 module with the textual-aware state fusion (TASF) module [24], achieving local neighborhood connectivity in state spaces with linear complexity. For details of the TASF, refer to the spatial-Mamba work [24]. The entire computing process of TAM can be formalized through three key equations: the state transition equation, the TASF equation, and the observation equation, which are expressed as

$$\begin{aligned}x_t &= \overline{\mathbf{M}}_t x_{t-1} + \overline{\mathbf{N}}_t u_t\\ h_t &= \sum_{k\in\Omega} \alpha_k x_{\rho_k(t)}\\ y_t &= \mathbf{P}_t h_t + \mathbf{Q} u_t\end{aligned} \tag{5}$$

where x_t represents the original state variable, h_t denotes the structure-aware state variable, Ω is the neighborhood set, α_k corresponds to a learnable weighting coefficient, and $\rho_k(t)$ specifies the index of the kth neighboring position relative to t. The TASF effectively encodes local dependencies into a refined state representation h_t, thereby enriching contextual information. By leveraging the TASF mechanism, the model can capture local structural dependencies in visual representations while preserving the computational efficiency of the original Mamba framework. In practical implementation, multiscale dilated convolutions [39] are employed to facilitate the linear combination of neighboring state variables, thereby enhancing spatial feature interactions and enabling efficient skip connections. Specifically, we follow spatial-Mamba [24] that utilizes three parallel dilated convolutions with dilation rates set as 1, 3, and 5, respectively. We conducted ablation studies in Section IV-E7 to validate the rationale behind the selection of ablation rate.

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b) TAB: Effective CD fundamentally relies on robust global scene understanding to achieve precise localization of changed areas. While Mamba's sequential processing paradigm excels in temporal data modeling, its inherent autoregressive nature constrains the receptive field that vision tasks typically demand. Thus, to expand the receptive field and enhance its ability to capture long-range spatial connections across the entire feature space, we introduce the transposed attention mechanism to our HAM. The SA mechanism constitutes the primary source of computational overhead in Transformer-based architectures. The resource demands of standard SA [23], [40] grow quadratically with respect to the number of input pixels, creating bottlenecks for high-resolution processing. To mitigate this issue, we propose the TAB, as illustrated in Fig. 2(d). By shifting the attention computation from spatial to channel dimensions, TAB achieves linear scaling in both memory and computation relative to input resolution. Specifically, TAB models global dependencies by constructing a cross-channel covariance map, thereby implicitly capturing long-range contextual information without incurring the prohibitive cost of full spatial attention. A key design element of TAB is the use of depth-wise convolutions to incorporate localized spatial information before the computation of global feature correlations. This mechanism ensures that the resulting attention map remains sensitive to spatial structures, thus enabling a better balance between local detail preservation and global context modeling. Given a input tensor $\mathbf{Y} \in \mathbb{R}^{\hat{H}\times\hat{W}\times\hat{C}}$, TAB first derives the *query* ($\mathbf{Q}$), *key* ($\mathbf{K}$), and *value* ($\mathbf{V}$) matrices, which is implemented through successive applications of pointwise ($W_p^{(\cdot)}$) and depthwise ($W_d^{(\cdot)}$) convolutional operators. This can be denoted as

$$\begin{aligned}\mathbf{Q} &= W_d^Q W_p^Q \mathbf{Y}\\ \mathbf{K} &= W_d^K W_p^K \mathbf{Y}\\ \mathbf{V} &= W_d^V W_p^V \mathbf{Y}.\end{aligned} \tag{6}$$

All convolutional layers are implemented without bias to streamline the computational flow. Subsequently, the $\mathbf{Q}$ and $\mathbf{K}$ tensors are reshaped to enable their dot-product to generate a compact transposed attention matrix $\mathbf{A} \in \mathbb{R}^{\hat{C}\times\hat{C}}$, significantly reducing the complexity compared to conventional attention matrices of size $\mathbb{R}^{\hat{H}\hat{W}\times\hat{H}\hat{W}}$ [23], [40]. The overall TAB operation is defined as

$$\begin{aligned}\hat{\mathbf{X}} &= W_p\,\mathrm{Attn}\left(\hat{\mathbf{Q}},\hat{\mathbf{K}},\hat{\mathbf{V}}\right)+\mathbf{X}\\ \mathrm{Attn}\left(\hat{\mathbf{Q}},\hat{\mathbf{K}},\hat{\mathbf{V}}\right) &= \hat{\mathbf{V}}\cdot\mathrm{Softmax}\left(\hat{\mathbf{K}}\cdot\hat{\mathbf{Q}}/\alpha\right)\end{aligned} \tag{7}$$

where $\mathbf{X}$ and $\hat{\mathbf{X}}$ denote the input and the transformed output feature maps, respectively. Here, $\hat{\mathbf{Q}} \in \mathbb{R}^{\hat{H}\hat{W}\times\hat{C}}$, $\hat{\mathbf{K}} \in \mathbb{R}^{\hat{C}\times\hat{H}\hat{W}}$, and $\hat{\mathbf{V}} \in \mathbb{R}^{\hat{H}\hat{W}\times\hat{C}}$ are the reshaped forms of the original projections. The scalar α is a learnable parameter that scales the key–query interaction before Softmax normalization. Following the design principles of standard multihead attention [23], we further divide the channel dimension into multiple heads, enabling the parallel computation of attention across different subspaces, thereby improving the model's capacity to learn diverse representations efficiently.

2) ICSF Block: Multistage feature fusion is critical for CD as it enables the progressive integration of complementary hierarchical representations, ensuring both fine-grained discriminability for subtle changes and robust contextual awareness for semantic-level transitions. Thus, we design the ICSF block to adaptively fuse cross-stage features, enhancing representation robustness through integrated channel–spatial interactions. The structure of the ICSF block is depicted in Fig. 3. This block enhances and integrates pretime and posttime feature representations by jointly modeling their semantic differences along both channel and spatial dimensions, using the channel-wise feature enhancement module (CFEM) and the spatial-wise feature enhancement module (SFEM), respectively. We expect that the helpful information is adaptively selected and merged along both spatial and channel dimensions. Given two input feature maps $\mathbf{F}_1, \mathbf{F}_2 \in \mathbb{R}^{C\times H\times W}$, the proposed ICSF block performs a fusion process and outputs a fused representation $\mathbf{F}_{\text{fused}} \in \mathbb{R}^{C\times H\times W}$ that incorporates complementary information from both inputs. The input features are first concatenated along the channel dimension. Then, to reduce dimensionality and enhance representation, $\mathbf{F}_{\text{cat}}$ is projected via a point-wise convolution (PWC) followed by a batch norm layer and a RELU activation layer. This process can be formulated as

$$\begin{aligned}\mathbf{F}_{\text{cat}} &= \mathrm{Concat}\left(\mathbf{F}_1,\mathbf{F}_2\right)\\ \mathbf{F}_{\text{proj}} &= \mathrm{ReLU}\left(\mathrm{BN}\left(\mathrm{PWC}\left(\mathbf{F}_{\text{cat}}\right)\right)\right).\end{aligned} \tag{8}$$

To emphasize discriminative channel features across the two inputs, we design a CFEM based on the squeeze-and-excitation (SE) style channel attention mechanism [41]. Specifically, global average pooling (GAP) is applied to obtain the channel descriptor. This descriptor is passed through two convolutions with ReLU and Sigmoid activations to generate channel-wise attention weights. The two convolution with kernel size 1×1 to perform channel-wise compression and expansion, respectively,

$$\begin{aligned}\mathbf{z}_c &= \mathrm{GAP}\left(\mathbf{F}_{\text{proj}}\right)\\ \mathbf{w}_c &= \sigma\left(\mathrm{PWC}\cdot\delta\left(\mathrm{DWC}\cdot\mathbf{z}_c\right)\right)\\ \mathbf{F}_c &= \mathbf{F}_{\text{proj}}\odot\mathbf{w}_c\end{aligned} \tag{9}$$

where $\delta(\cdot)$ and $\sigma(\cdot)$ denote ReLU and Sigmoid functions, respectively. The weights are used to recalibrate the projected features. To capture structural discrepancies and enhance spatial-wise feature representations, we design an SFEM. The SFEM applies a 3×3 depthwise convolution (DWC) followed by a batch normalization (BN) layer and a ReLU activation. This operation focuses on spatial interaction while maintaining computational efficiency. This process can be represented as

$$\mathbf{F}_s = \mathrm{ReLU}\left(\mathrm{DWC}_{3\times3}\left(\mathbf{F}_{\text{proj}}\right)\right). \tag{10}$$

The channel- and spatial-enhanced features are aggregated and passed through a final 1×1 point-wise convolution and BN layer. Besides, a residual connection is then applied to stabilize the training

$$\begin{aligned}\mathbf{F}_{\text{agg}} &= \mathrm{BN}\left(\mathrm{PWC}\left(\mathbf{F}_c+\mathbf{F}_s\right)\right)\\ \mathbf{F}_{\text{fused}} &= \mathrm{ReLU}\left(\mathbf{F}_{\text{agg}}+\mathbf{F}_{\text{proj}}\right).\end{aligned} \tag{11}$$

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In all, the proposed ICSF block integrates useful information from two sources by leveraging an interactive channel–spatial modeling. This design ensures that both high-level semantics and fine-grained spatial cues are preserved and enhanced during fusion, providing robust representations for the binary CD. In future work, we will continue to explore potential improvements to the TAM to further enhance the perception of local details and therefore achieve superior CD performance.

D. Optimization Objectives

Given that this study aims to investigate the applicability of the HAM architecture for CD tasks, we employ standard loss functions commonly used in the field for network optimization. Notably, we do not incorporate additional accuracy-enhancing techniques. Despite this, as demonstrated in the experimental section, our proposed architecture achieves highly competitive and even SOTA performance on the CD task. Since binary CD can be formally expressed as a constrained semantic segmentation task [42], we optimize the network using a standard cross-entropy loss function

$$\mathcal{L}_{ce} = -\frac{1}{N_{train}} \sum_{i=1}^{N_{train}} \sum_{c=0}^{1} \tilde{\mathbf{Y}}_i(c) \log\left(\mathbf{P}_i(c)\right) \tag{12}$$

where $\tilde{\mathbf{Y}}_i$ denotes the encoded form of the ground truth (GT) $\mathbf{Y}_i$ and $\mathbf{P}_i = \mathcal{F}_\theta(\mathbf{X}_i^{\mathrm{T}_1}, \mathbf{X}_i^{\mathrm{T}_2})$ denotes the Softmax-normalized output of the binary change detector. To mitigate the class imbalance between changed and unchanged pixels, we further introduce the Lovász-Softmax loss [43]. The final loss function is thus defined as

$$\mathcal{L}_{final} = \mathcal{L}_{ce} + \mathcal{L}_{lov}. \tag{13}$$

This combined loss formulation ensures both accurate classification and improved robustness against class imbalance.

IV. Experiments and Discussions

A. Datasets

1) LEVIR-CD+ [44]: This CD dataset consists of 985 pairs of very high-resolution images (0.5 m per pixel), each with a spatial dimension of 1024×1024 pixels. These multitemporal images show substantial changes in building construction. The dataset covers a wide spectrum of building types, including villa residences, high-rise apartments, small garages, and large warehouses, thereby providing a comprehensive representation of both newly constructed buildings and demolished structures.

2) WHU-CD [45]: This dataset is specifically curated for the building CD task. It consists of two aerial image datasets captured at different times and cited with a spatial resolution of 0.3 m per pixel. This dataset focuses on detecting changes in large and sparsely distributed buildings. The 2012 dataset covers an area of 20.5 km^2, containing 12 796 buildings, whereas the 2016 dataset, covering the same area, records an increase to 16 077 buildings, reflecting significant urban expansion over four years. We adhere to the official dataset protocol [45] to partition the dataset into training and testing subsets for experimental evaluation.

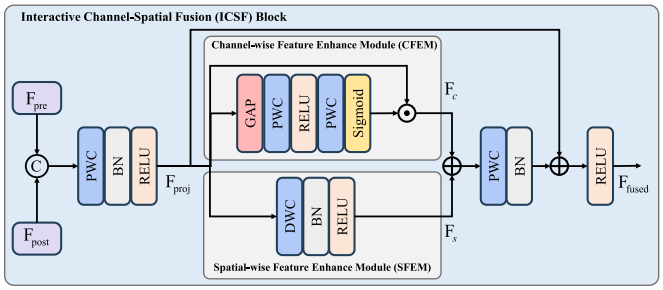


Fig. 3. Structure of the ICSF block. It adaptively fuses cross-stage features, enhancing robust spatial–temporal feature representation by jointly modeling their semantic differences in both channel- and spatial-wise dimensions.

3) SYSU-CD [46]: This dataset is a category-agnostic CD dataset comprising 20 000 pairs of aerial images, and the spatial resolution of the images is 0.5 m per pixel. This dataset is characterized by urban and coastal transformations, including the construction of high-rise buildings and the expansion of the infrastructure. It encompasses diverse change scenarios such as urban development, suburban expansion, groundwork modifications, vegetation alterations, road widening, and coastal construction.

B. Experimental Setup

The proposed architectures are implemented in the PyTorch framework. All training and inference experiments were conducted on NVIDIA RTX 3090 GPUs. Three datasets undergo uniform preprocessing, where multitemporal image pairs and their corresponding GT masks are systematically cropped into 256×256 pixel patches to ensure consistent network input dimensions. The trained networks are then applied to the test set to infer results on images of their original sizes. The attention heads and channel expansion factor in TAM are set as 8 and 2.66, respectively. The network is trained using the AdamW optimizer [47] with a base learning rate of 1×10^{-4} and weight decay coefficient 5×10^{-3}. The batch size is configured as 16. The training iterations is set as 5 00 000 for all three datasets. We employ data augmentation techniques including random rotation, left–right flipping, and top–bottom flipping.

C. Evaluation Metrics

To assess the performance of different comparison methods, we employ a set of evaluation metrics tailored to the specific requirements of binary CD. We adopt five commonly used metrics: recall (abbreviated as Rec), precision (abbreviated as Prec), general accuracy (OA), $F1$-score ($F1$), and intersection over union (IoU). Higher values indicate better performance for all six metrics. The mathematical formulations of these metrics can be found in [42], [48], and [49].

D. Comparison Results on Benchmark Datasets

To thoroughly evaluate the effectiveness of our HAM-CD on binary CD, we conduct comparative experiments with a set of representative approaches built upon CNN,

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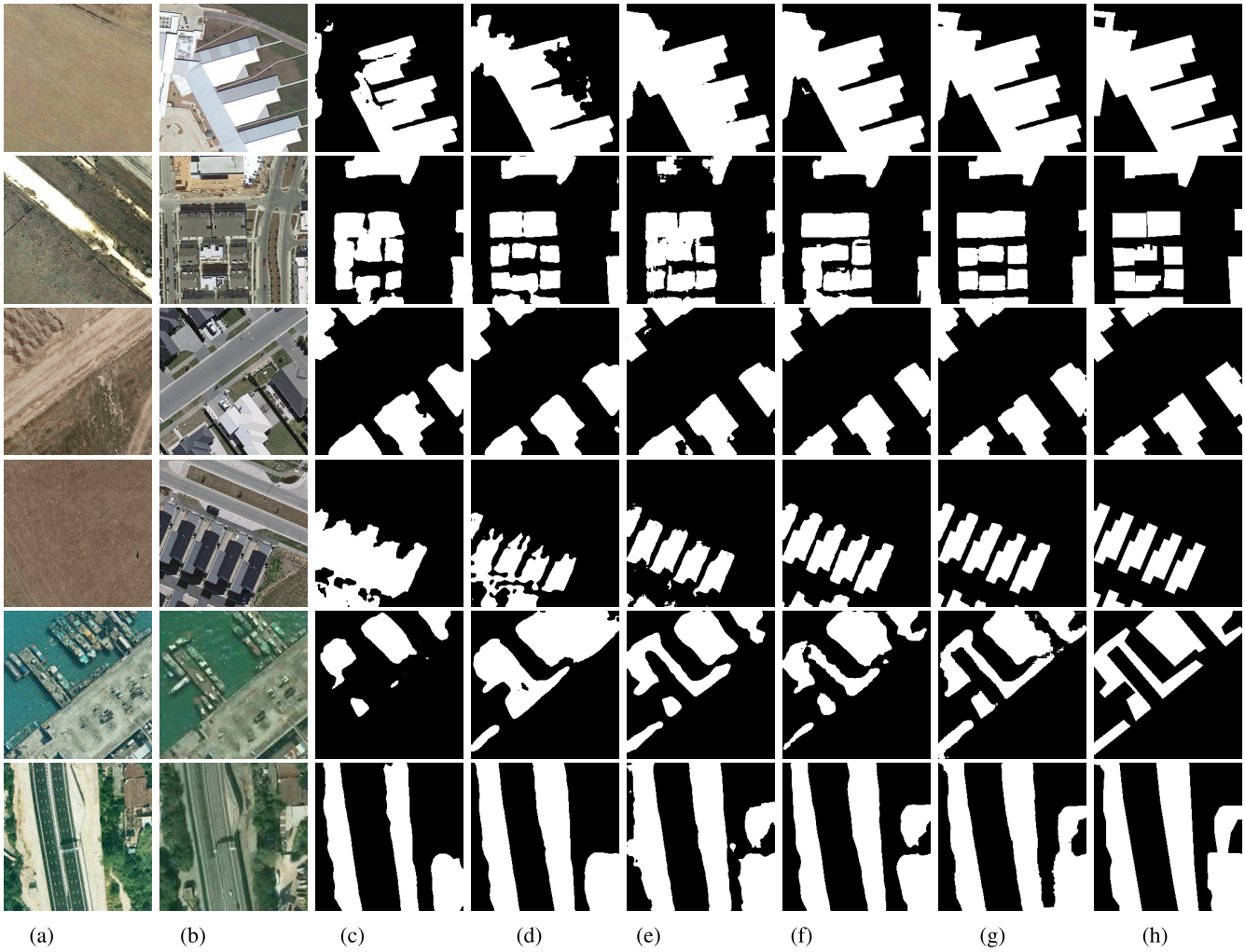


Fig. 4. Qualitative comparison results obtained by different CD methods on the three BCD datasets. From the top to the bottom row, we demonstrate the detected change maps generated by different methods on the LEVIR-CD, WHU-CD, and SYSU-CD datasets, respectively. (a) Pretime. (b) Posttime. (c) SEIFNet. (d) BIT. (e) Cha-former. (f) MambaBCD. (g) Ours. (h) GT.

Transformer, and Mamba architectures for comprehensive comparison. The CNN-based methods include FC-Siam-Diff [12], FC-Siam-Conc [12], and SiamCRNN [30], which utilize ResNet-18 [57] and ResNet-50 [57] backbones. Additional CNN-based baselines include DSIFN [50], SNUNet [31], HANet [51], CGNet [52], SEIFNet [53], and DDPM-CD [54]. For Transformer-based models, we evaluate against ChangeFormer (versions 1 and 6) [13], BIT [17] (with ResNet-50 as feature extractors), TransUNetCD [19], SwinSUNet [15], CTDFormer [16], M-Swin [55], and RFormer [56]. For the Mamba-based method, we compare our HAM-CD with the outstanding MambaBCD [22] method. To ensure a fair comparison, we have adopted the following experimental comparison strategy. For methods without publicly available source code, we have adopted the performance metrics reported in their original publications. For approaches with accessible implementations, we have retrained and evaluated the models on the relevant datasets using the hyperparameter configurations specified in their respective articles. Besides, we have applied the same loss function and data augmentation strategies across all methods, including our proposed approach. This process helps mitigate potential biases that may arise from inconsistent experimental conditions. Furthermore, in Tables I and II, CNN-based, Transformer-based, Mamba-based, and HAM models are denoted by **C**, **T**, **M**, and **H**, respectively. The quantitative results of our proposed HAM-CD architecture and the competing methods on the three benchmark datasets are summarized in Tables I and II. Our models consistently surpass other baselines across all evaluated metrics. In particular, HAM-CD attains the best performances regarding OA, $F1$, and IoU on all three evaluated benchmark datasets, highlighting the effectiveness of the HAM block for binary CD. Our HAM-CD consistently surpasses the current SOTA method, Mamba-CD, across all three benchmark datasets, owing to the enhanced local pixel relationship modeling capability of the proposed HAM block. Furthermore, visual comparisons in Fig. 4 depict change maps detected by different comparison methods on the three benchmark test datasets. Compared with other methods, the proposed method is capable of detecting more accurate change maps, accurately matching the GT change maps. For example, a comparative analysis between Fig. 4(f) and (g) demonstrates

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TABLE I

QUANTITATIVE COMPARISON RESULTS ON THE LEVIR-CD [44] AND WHU-CD [45] DATASETS. FOR THIS TABLE, AS WELL AS THE REST OF THE TABLES, THE BEST PERFORMANCES ARE EMPHASIZED IN BOLD, AND THE SECOND-BEST PERFORMANCES ARE UNDERLINED

Type	Method	LEVIR-CD [44]					WHU-CD [45]				
		Rec	Prec	OA	F1	IoU	Rec	Prec	OA	F1	IoU
C	FC-EF [12]	71.77	69.12	97.54	70.42	54.34	86.33	83.50	98.87	84.89	73.74
	FC-Siam-Diff$_{18}$ [12]	74.02	81.49	98.26	77.57	63.36	84.69	90.86	99.13	87.67	78.04
	FC-Siam-Conc$_{18}$ [12]	78.49	78.39	98.24	78.44	64.53	87.72	84.02	98.94	85.83	75.18
	SiamCRNN-18$_{20}$ [30]	84.25	81.22	98.56	82.71	70.52	90.48	91.56	99.34	91.02	83.51
	SiamCRNN-50$_{20}$ [30]	81.61	85.39	98.68	83.46	71.61	91.45	86.70	99.30	90.57	82.76
	DSIFN$_{20}$ [50]	84.36	83.78	98.70	84.07	72.52	83.45	**97.46**	99.31	89.91	81.67
	SNUNet$_{22}$ [50]	84.36	83.78	98.70	84.07	72.52	87.36	88.04	99.10	87.70	78.09
	HANet$_{23}$ [51]	75.53	79.70	98.22	77.56	63.34	88.30	88.01	99.16	88.16	78.82
	CGNet$_{23}$ [52]	86.02	81.46	98.63	83.68	71.94	90.79	94.47	99.48	92.59	86.21
	SEIFNet$_{24}$ [53]	81.86	84.83	98.66	83.32	71.41	90.66	91.93	99.36	91.29	83.98
	DDPM-CD$_{25}$ [54]	89.87	91.43	99.09	90.91	83.35	90.82	94.69	99.42	92.65	86.31
T	ChangeFormerV1$_{22}$ [13]	77.00	82.18	98.38	79.51	65.98	84.30	90.80	99.11	87.43	77.67
	ChangeFormerV6$_{22}$ [13]	78.57	82.97	98.60	80.71	66.12	81.90	85.49	98.83	83.66	71.91
	BIT-50$_{22}$ [17]	77.90	81.42	90.60	79.62	66.14	90.33	89.70	99.26	90.01	81.84
	TransUNetCD$_{22}$ [19]	84.18	83.08	98.66	83.63	71.86	90.50	85.48	99.09	87.79	78.44
	SwinSUNet$_{22}$ [15]	85.85	85.34	98.92	85.60	74.82	92.03	94.08	99.50	93.04	87.00
	CTDFormer$_{23}$ [16]	80.03	80.58	98.40	80.30	67.09	85.37	92.23	99.20	88.67	79.65
	M-Swin$_{24}$ [55]	90.05	92.07	99.04	91.05	84.37	91.37	94.55	99.42	92.93	86.09
	RFormer$_{24}$ [56]	89.77	92.16	99.09	90.95	83.30	88.65	93.89	99.32	91.92	83.82
M	MambaBCD-Small$_{24}$ [22]	86.49	89.17	99.02	87.81	78.27	92.29	95.90	99.57	94.06	88.79
	MambaBCD-Base$_{24}$ [22]	87.57	89.24	99.06	88.39	79.20	92.23	96.18	99.58	94.19	89.02
H	HAM-CD	**90.53**	**92.57**	**99.15**	**91.54**	**84.40**	**94.14**	95.36	**99.58**	**94.74**	**90.02**

that the CD maps generated by HAM-CD exhibit fewer artifacts, more precise boundaries, and higher consistency with the GT. This demonstrates that the incorporation of HAM and ICSF modules enhances the original Mamba architecture's capability in local pixel relationship modeling and feature representation, enabling our hybrid model HAM-CD to generate change maps with more accurate edge delineation and finer detail preservation, complementing the quantitative improvements in $F1$ and IoU. Furthermore, comprehensive qualitative comparisons on three datasets demonstrate that HAM-CD consistently achieves precise detection of diverse change patterns across varying categories, scales, and quantities.

Table II also compares the computational complexity of various binary CD architectures, evaluated on a standard NVIDIA RTX 3090 GPU with a typical input size 256 $\times$ 256 on the SYSU dataset. Leveraging the linear-time complexity of the TAM and the TAB, the proposed HAM-CD records only 25.17 GFLOPs, representing a significant reduction in computational burden compared to most CNN and Transformer-based counterparts. Compared to lightweight CNN-based methods such as SNUNet [31] and SiamCRNN [30], HAM-CD reduces GFLOPs by approximately 85.73% and 70.95%, while simultaneously yielding a substantial 10.29% and 2.81% absolute improvement in $F1$-score on the SYSU-CD dataset. Although HAM-CD increases GFLOPs by 78.13% relative to the most lightweight CNN-based baseline, FC-EF [12], it yields a large performance gain, elevating the absolute $F1$-score by 7.62%. This indicates that the modest computational increment provides a substantial boost in modeling capacity, far surpassing the efficiency of early lightweight CNN architectures. Besides, when compared to typical lightweight Transformer-based methods like ChangeFormerV1 [13] and SwinSUNet [15], HAM-CD reduces GFLOPs by approximately 46.01% and 42.14%, while achieving a 5.74% and 1.85% absolute improvement in $F1$-score. Finally, when compared with MambaBCD [22], although our HAM-CD method slightly exceeds MambaBCD-Small in the number of parameters, the GFLOPs of the HAM-CD account for only 21.9% of MambaBCD-Small. This drastic reduction in computational cost demonstrates that HAM-CD effectively overcomes the quadratic complexity bottleneck of Transformers. Such a favorable tradeoff, where high precision is maintained under a lightweight computational budget, makes HAM-CD exceptionally suitable for deployment on resource-constrained edge devices for real-time remote sensing tasks. In terms of inference speed, it can be observed from Table II that our method achieves faster inference speed compared to typical Transformer-based approaches such as ChangeFormer [13] and Mamba-based methods like MambaBCD [22], which further demonstrates its greater suitability for deployment in practical remote sensing applications. To provide a more intuitive illustration of the tradeoff between the performance gains and computational costs, Fig. 1 illustrates the relationship between GFLOPs and the $F1$ for CD. As demonstrated, the proposed method attains an optimal balance between model performance and efficiency.

E. Ablation Studies

1) Effectiveness of HAM and ICSF: We perform a series of experiments to validate the effectiveness of the proposed HAM and ICSF. The quantitative results on the LEVIR-CD [44] dataset for binary CD are summarized in Table III. In the baseline model, the HAM block is replaced with the Vision Mamba block [35] and the ICSF block is replaced

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TABLE II

EVALUATION OF VARIOUS CD MODELS ON THE SYSU-CD DATASET [46], ALONG WITH AN ANALYSIS OF THEIR MODEL COMPLEXITIES

Type	Method	Model Complexity			SYSU-CD [45]				
		Param (M)	GFLOPs	Inference Speed (s)	Rec	Prec	OA	F1	IoU
C	FC-EF [12]	**1.35**	**14.13**	**0.01**	75.17	76.47	88.69	75.81	61.04
	FC-Siam-Diff$_{18}$ [12]	1.35	18.66	0.01	75.30	76.28	88.65	75.79	61.01
	FC-Siam-Conc$_{18}$ [12]	1.54	21.07	0.01	76.75	73.67	88.05	75.18	60.23
	SiamCRNN-18$_{20}$ [30]	18.85	86.64	0.11	76.83	84.80	91.29	80.62	67.54
	SiamCRNN-50$_{20}$ [30]	44.45	185.30	0.27	78.40	83.41	91.23	80.83	67.82
	DSIFN$_{20}$ [50]	35.73	329.03	0.22	**82.02**	75.83	89.59	78.80	65.02
	SNUNet$_{22}$ [31]	10.21	176.36	0.06	72.21	74.09	87.49	73.14	57.66
	HANet$_{23}$ [51]	2.61	70.68	0.02	76.14	78.71	89.52	77.41	63.14
	CGNet$_{23}$ [52]	33.68	329.58	0.21	74.37	86.37	91.19	79.92	66.55
	SEIFNet$_{24}$ [53]	39.08	167.75	0.24	78.29	78.61	89.86	78.45	64.54
	DDPM-CD$_{25}$ [54]	372.93	405.75	2.27	79.95	86.19	91.22	80.95	66.69
T	ChangeFormerV1$_{22}$ [13]	29.84	46.62	0.20	75.82	79.65	89.73	77.69	63.52
	ChangeFormerV6$_{22}$ [13]	41.03	811.15	0.25	72.38	81.70	89.67	76.76	62.29
	BIT-50$_{22}$ [17]	24.28	224.61	0.15	81.84	85.02	98.67	83.40	71.53
	TransUNetCD$_{22}$ [19]	28.37	244.54	0.17	77.73	82.59	90.88	80.09	66.79
	SwinSUNet$_{22}$ [15]	39.28	43.50	0.24	79.75	83.50	91.51	81.58	68.89
	CTDFormer$_{23}$ [16]	3.85	303.77	0.02	75.53	80.80	90.00	78.08	64.04
	M-Swin$_{24}$ [55]	57.69	122.13	0.35	78.85	87.27	91.83	83.05	69.45
	RFormer$_{24}$ [56]	67.42	142.73	0.41	79.37	86.36	91.57	82.64	69.12
M	MambaBCD-Small$_{24}$ [22]	49.94	114.82	0.25	78.25	**87.99**	92.35	82.83	70.70
	MambaBCD-Base$_{24}$ [22]	84.70	179.32	0.37	80.31	86.11	92.30	83.11	71.10
H	HAM-CD	51.26	25.17	0.19	79.37	87.92	**92.56**	**83.43**	**71.57**

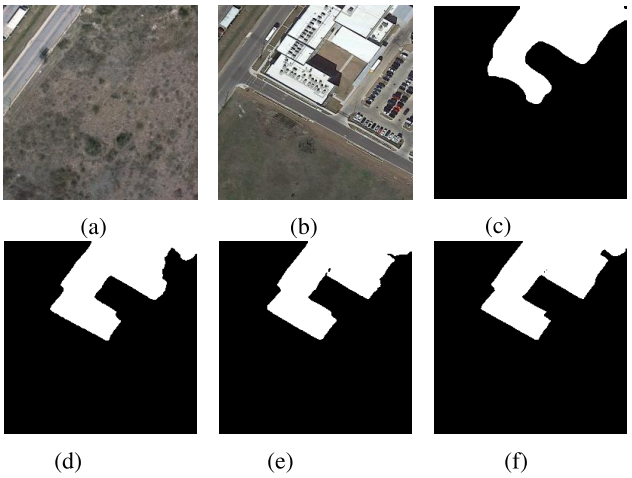


Fig. 5. Visualization of ablation results of major components. (a) and (b) Pretime and posttime images, respectively. (c) Change map detected by the baseline model. (d) and (e) Change maps detected by adding the ICSF module and the HAM block into the baseline model. The change map detected by our full model is shown in (f), which exhibits finer details.

with a convolution block. Compared to the baseline model, HAM and ICSF individually yield improvements of 3.09 and 2.23 in $F1$ score, respectively. When combined, the two modules contribute a 4.43 performance gain in $F1$-score. We visualize the effectiveness of major components in Fig. 5. As shown, both the HAM block and the ICSF module contribute significantly to refining the textual details of the detected change map. Furthermore, the change map detected by our full-setting model, which integrates both HAM and ICSF along with other optimized components, exhibits superior performance in detecting precise boundaries and subtle changes compared to baseline methods. The ablation results confirm the complementary effectiveness of HAM and ICSF in obtaining accurate CD and improved model performance. The HAM block enhances the global and local feature representation ability, while ICSF's feature aggregation mechanism enables refined spatial–temporal cross-stage feature fusion. Their integration addresses both temporal and spatial dependencies, which is critical for accurate CD.

TABLE III

ABLATION STUDY OF THE PROPOSED HAM AND ICSF ON THE LEVIR-CD [44] DATASET

Basline	ICSF	HAM	Rec	Prec	OA	F1	IoU
✓	×	×	86.11	88.09	98.50	87.11	80.31
✓	✓	×	88.36	90.35	98.80	89.34	82.37
✓	×	✓	89.21	91.23	99.01	90.20	83.17
✓	✓	✓	**90.53**	**92.57**	**99.15**	**91.54**	**84.40**

2) Effectiveness of Tab in HAM: We perform an investigation to evaluate the influence of different SA configurations, with the corresponding results summarized in Table IV. For a fair comparison, we integrate the global multihead SA (G-MSA) [40] mechanism, which is commonly employed in prior CNN–Transformer hybrid architectures, into each HAM block. We conduct systematic ablation studies by replacing our TAB in each HAM block with three different

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TABLE IV
EVALUATION OF VARIED SA SCHEMES ON THE LEVIR-CD [44] DATASET

Attentions	Param	GFLOPs	Rec	Prec	OA	F1	IoU
G-MSA [40]	74.90	131.36	87.28	89.25	98.60	88.25	81.37
W-MSA [58]	63.31	96.87	87.96	89.93	98.90	88.94	82.01
Spatial [59]	58.93	82.98	89.34	91.31	98.56	90.33	83.25
TAB (Ours)	51.26	25.17	**90.53**	**92.57**	**99.15**	**91.54**	**84.40**

TABLE V
ABLATION STUDY OF THE CONSTRUCTION STRATEGIES OF TAM AND TRANSPOSED SA IN HAB ON THE LEVIR-CD [44] DATASET

Strategies	Rec	Prec	OA	F1	IoU
Serial (TAB → TAM)	88.68	90.81	98.81	89.73	82.47
Serial (TAM → TAB)	88.89	90.93	99.02	89.94	82.85
Parallel (Ours)	**90.53**	**92.57**	**99.15**	**91.54**	**84.40**

TABLE VI
COMPARISON OF DIFFERENT OPERATIONS TO FUSE FEATURES OF THE TWO BRANCHES IN HAM

Operations	Rec	Prec	OA	F1	IoU
Summation	89.98	92.02	99.11	90.89	83.72
Concatenation	**90.53**	**92.57**	**99.15**	**91.54**	**84.40**

attention mechanisms, the widely used G-MSA, window-based SA (W-MSA), and spatial attention [59]. To mitigate memory overhead, input feature maps for G-MSA are uniformly downsampled to one-fourth of their original spatial resolution. As observed in Table IV, compared to the original G-MSA and the W-MSA employed in the Swin Transformer [58], our TAB yields superior quantitative performance by a large margin. Besides, our TAB outperforming the spatial attention mechanism [59] by 1.19, 1.22, and 1.26 in recall, precision, and $F1$-score, respectively. Furthermore, our TAB simultaneously reduces the computational cost by 7.67 and 57.81 GFLOPs. These findings highlight the efficiency and effectiveness of the TAB design in balancing performance and complexity.

3) Effectiveness of Different Hybrid Patterns in HAM: A series of ablation experiments is performed to investigate the impact of different hybrid patterns of the HAB module. The results are presented in Table V. As shown, the parallel construction strategy achieves a substantial performance enhancement over the serial counterparts, demonstrating a significant $F1$-score improvement of 2.60 compared to serial (TAM → TAB). This performance gap can be attributed to a fundamental difference in architectural design between serial and parallel fusion strategies. In serial fusion, the secondary module operates on a filtered feature space produced by the predecessor, which is constrained by the inductive bias of the first module. In contrast to serial architectures, our proposed parallel integration can effectively prevent the filtering effect and the cascading loss of fine-grained information that often occurs during sequential processing. The HAM block allows both TAM and TAB to operate simultaneously on the same unfiltered input features. Furthermore, this parallel structure facilitates a balanced and flexible fusion process. It enables the model to learn dynamic weightings between global and local representations at every layer. Thus, this parallel integration strategy enables the HAM module to fully leverage the global contextual modeling capabilities of TAB and the fine-grained local pixel relationship learning power of TAM. Consequently, it establishes a more robust global–local modeling capability. Besides, we also carried out experiments to examine the effects of integrating the outputs of TAM and TAB of HAM and report the ablation results in Table VI. We evaluate two different methods for fusing the output features: summation and concatenation. As shown, the concatenation operation achieves better performance. We hypothesize that concatenation outperforms addition because it maintains richer feature hierarchies, allowing the model to explicitly leverage both shared and unique patterns across temporal phases. In addition, in CD tasks, directly summing feature maps from different temporal phases may suppress subtle changes due to dominant background signals, whereas concatenation followed by adaptive weighted fusion through subsequent convolution layers can more flexibly preserve critical discrepancies. Thus, concatenation facilitates the learning of complex feature relationships, demonstrating superior performance in tasks requiring fine-grained analysis.

TABLE VII
COMPARISON OF DIFFERENT FUSION MODULES OF CROSS-STAGE FEATURES ON THE LEVIR-CD [44] DATASET

Fusion Type	Param	GFLOPs	Rec	Prec	OA	F1	IoU
Summation	**48.76**	**24.55**	87.74	89.76	88.92	88.74	81.21
Concatenation	48.98	24.60	87.95	89.98	97.96	88.96	81.42
Convolution	49.38	24.66	89.21	91.23	99.01	90.20	83.17
SKFF [60]	52.63	25.42	87.96	89.94	98.85	88.95	82.03
Residual [22]	50.15	24.98	89.32	91.36	99.07	90.34	83.27
ICSF w/o CA	50.38	25.02	89.98	92.04	99.10	90.79	83.94
ICSF (Ours)	51.26	25.17	**90.53**	**92.57**	**99.15**	**91.54**	**84.40**

4) Comparison of Different Fusion Modules for Cross-Stage Spatial–Temporal Features: We compare different fusion modules for cross-stage spatial–temporal feature integration, including element-wise summation, concatenation, a standard convolution layer, the selective kernel feature fusion (SKFF) module [60], the residual fusion block applied in ChangeMamba [22], and our ICSF with channel attention mechanism removed. We summarize the comparison results in Table VII, from which we can conclude that our ICSF achieves superior performance. Compared to the SKFF module, which is specifically designed for feature fusion, our ICSF achieves 2.59 gains in the $F1$-score. Our ICSF outperforms the residual fusion strategy applied in ChangeMamba [22] by 1.20 in the $F1$-score. The superior performance obtained by our ICSF module is attributed to the pixel-wise adaptive selection enabled by the channel attention and depth-wise convolution. Regarding computational overhead, the proposed ICSF module introduces only a marginal increase in both parameter count and computational complexity, with

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TABLE VIII

EVALUATION FOR THE SUBCOMPONENTS OF ICSF ON THE LEVIR-CD [44] DATASET

Baseline	SFEM	CFEM	Rec	Prec	OA	F1	IoU
✓	×	×	88.91	90.92	99.00	89.92	82.66
✓	✓	×	89.73	91.75	99.03	90.75	83.16
✓	×	✓	90.03	92.02	99.08	91.04	83.76
✓	✓	✓	**90.53**	**92.57**	**99.15**	**91.54**	**84.40**

TABLE IX

EVALUATION OF VARIED INPUT SIZE ON THE LEVIR-CD [44] DATASET

Input Size	Param	GFLOPs	Rec	Prec	OA	F1	IoU
128 × 128	12.81	6.29	84.12	86.00	98.50	85.06	74.01
192 × 192	12.81	6.29	88.59	90.58	98.95	89.58	81.83
256 × 256	51.26	25.17	**90.53**	**92.57**	**99.15**	**91.54**	**84.40**
320 × 320	115.33	56.63	89.37	91.39	99.03	90.38	82.45
384 × 384	205.04	100.68	87.58	89.56	98.85	89.57	79.49

TABLE X

EVALUATION OF VARIED STAGES OF THE DECODER ON THE LEVIR-CD [44] DATASET

Stages	Param	GFLOPs	Rec	Prec	OA	F1	IoU
2	25.63	12.58	85.78	87.65	97.24	86.70	79.98
3	38.44	18.88	88.65	90.23	98.95	89.44	82.47
4	51.26	25.17	**90.53**	**92.57**	**99.15**	**91.54**	**84.40**
5	58.07	38.46	90.36	92.25	98.79	91.28	84.09

approximately 5.1% and 2.5% more requirements compared to element-wise summation, respectively. All experimental results in Table VII demonstrate that ICSF achieves a favorable balance between added complexity and performance benefits, providing enhanced cross-stage feature interaction while maintaining practical computational efficiency.

5) Effectiveness of the Subcomponents ICSF: To validate the design rationality of ICSF, we performed a systematic analysis of its two major subcomponents—CFEM and SFEM—in a modular breakdown manner. The evaluation results on the LEVIR-CD [44] dataset are reported in Table VIII. The baseline model is established by removing both SFEM and CFEM. As shown in Table VIII, compared to the baseline, the SFEM and CFEM subcomponents individually enhance the $F1$-score by 0.83 and 1.12, respectively. Furthermore, their integration produces a markedly greater effect, achieving an overall performance gain of 1.62. The ablation results confirm the complementary effectiveness of SFEM and CFEM in achieving accurate CD. SFEM improves the model's ability to extract structural detail features, while CFEM enhances the discriminative ability of channel-wise feature representations. Their integration promotes the interaction between channel- and spatial-wise feature representations.

6) Effectiveness of Different Architectural Configurations: To further investigate the architectural design rationality, we conduct a series of controlled experiments by altering the network configuration to assess the impact of varying input image size and stages of the encoder. The evaluation results are summarized in Tables IX and X. By observing the experimental results, we can make several observations. First, larger input sizes lead to consistent improvements in model performance as the input spatial resolution increases from 128 × 128 to 256 × 256. However, as the input spatial resolution increases beyond 256 × 256, the performance diminishes slightly, while the number of parameters and computational cost increase substantially. Second, deeper stage configurations of the decoder can improve detection accuracy when the stage number increases from 2 to 4. However, a slight performance degradation is observed when increasing the number of stages from 4 to 5, suggesting that the model's capacity is already fully utilized when the stage number equals 4, representing an optimal configuration. In summary, this set of experiments collectively confirms that different architectural configurations are essential for striking a balance between performance and computational efficiency.

TABLE XI

EVALUATION OF DIFFERENT DILATION RATE COMBINATIONS FOR TAM ON THE LEVIR-CD [44] DATASET

Dilation Rates	Rec	Prec	OA	F1	IoU
(1, 2, 4)	89.28	91.35	98.87	90.36	82.44
(1, 3, 5)	**90.53**	**92.57**	**99.15**	**91.54**	**84.40**
(1, 4, 6)	90.39	92.14	99.05	91.22	84.38

TABLE XII

EVALUATION OF DIFFERENT ATTENTION HEADS NUMBER FOR TAB ON THE LEVIR-CD [44] DATASET

Head Number	Rec	Prec	OA	F1	IoU
2	89.10	90.84	97.42	89.96	82.37
4	89.32	91.21	98.15	90.26	83.75
6	89.93	91.78	98.52	90.85	84.07
8	**90.53**	**92.57**	**99.15**	**91.54**	**84.40**
10	90.38	92.25	99.06	91.33	84.39

7) Effectiveness of Dilation Combinations in TAM: In the internal design of TAM, we employ three parallel dilated convolutions [39] to facilitate the linear combination of neighboring state variables, thereby enhancing multiscale spatial feature interactions. In this section, we evaluate the effectiveness of different dilation rate combinations and report the results in Table XI. As shown, our HAM-CD achieves the best performance when the dilation rates of the three dilation convolutions are set as (1, 3, 5). When the dilation rates were increased to (1, 4, 6), a slight performance degradation was observed. We conjecture that HAM-CD had already saturated at the rates of (1, 3, 5). Therefore, larger rates did not yield further performance gains.

8) Evaluation of Attention Head Number in TAB: We systematically evaluate the impact of the number of attention heads in the TAB, with quantitative results presented in Table XII. The experimental results demonstrate that model performance does not monotonically improve with an increasing number of attention heads. When the number of heads is set to 2, our method achieves baseline performance. Our HAM-CD achieves the best performance when the head count

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TABLE XIII

EVALUATION OF DIFFERENT MAMBA-BASED ENCODERS OF HAM-CD ON THE LEVIR-CD [44] DATASET

Different Encoders	Rec	Prec	OA	F1	IoU
Vision Mamba [36]	89.83	90.95	97.82	90.38	82.76
Spatial-Mamba [24]	90.62	91.73	99.11	91.18	83.87
VMamba (Ours)	**90.53**	**92.57**	**99.15**	**91.54**	**84.40**

is set as 8, yielding the optimal $F1$-score of 91.54. It is noteworthy that when the number of heads further increases to 10, the $F1$-score exhibits a slight decline compared to the peak values. This phenomenon reveals that too few attention heads may limit the model's capacity to represent change information, while an excessive number of heads potentially introduces redundancy or leads to overfitting, consequently causing a minor performance degradation. Therefore, we employ 8 attention heads in the TAB to strike a balance between model complexity and efficacy for remote sensing CD.

9) Evaluation of Different Mamba-Based Encoders: To justify the selection of the encoder for our HAM-CD model, we have conducted an ablation study investigating the impact of various representative Mamba-based backbones. Specifically, we compared our VMamba [35] encoder with Vision Mamba [36] and spatial-Mamba [24] by substituting each into the encoder stage while keeping the HAM-CD decoder constant. To ensure a fair comparison, all evaluations are conducted using pretrained models of comparable parameters. As illustrated in Table XIII, the VMamba-based configuration consistently outperforms the other candidates across nearly all key metrics. A comparative analysis of the underlying mechanisms reveals the advantages of our chosen backbone. First, while Vision Mamba [36] provides a baseline for Mamba-based vision encoders, its performance is limited by its standard 1-D bidirectional scanning. This linear approach often fails to capture the intricate 2-D spatial relationships necessary for high-precision CD. Second, although spatial-Mamba [24] demonstrates competitive recall at 90.62%, VMamba [35] achieves a more robust balance across all evaluation metrics. This underscores the superior capability of VMamba [35] in extracting essential features from multitemporal images. We hypothesize that while spatial-Mamba emphasizes the modeling of spatial structures, this specialized focus potentially compromises its global contextual modeling capacity when employed as an image encoder, making it slightly weaker than that of VMamba [35] in extracting essential spatial–temporal features from multitemporal images for CD. In contrast, VMamba's multidirectional scanning ensures that each pixel aggregates long-range dependencies from the entire spatial context. Thus, we have chosen VMamba [35] as the encoder for our HAM-CD.

F. Cross-Domain Verification Under Degraded Scenes

To evaluate the robustness of our method under degraded input conditions, we deliberately introduce various types of degradations during the inference stage, including Gaussian noise, Gaussian blur, and varied input spatial resolutions. We compare our method with other representative methods on the LEVIR-CD+ dataset and use the $F1$-score as an evaluation metric. The comparison results are demonstrated in Tables XIV–XVI. The results presented in the three tables lead us to the following observations: first, in the case of Gaussian noise, all compared methods suffer considerable $F1$ degradation between 9.23% and 47.91%, except for Mamba-based methods, such as MambaBCD and our HAM-CD. When subjected to noise with a standard deviation of 20, our HAM-CD achieves more robust performance than MambaBCD, with a mere 5.45% reduction in $F1$-score. Second, similar trends are observed under Gaussian blur corruption. Especially, CNN-based models show significantly weak robustness. When the blur strength, measured by standard deviation, increases from 0 to 2.0, the $F1$-scores of both SiamCRNN-50 and DSIFN drop sharply by nearly 70%. In contrast, Transformer- and Mamba-based backbones such as

TABLE XIV

COMPARISON OF ROBUSTNESS AGAINST DIFFERENT LEVELS OF GAUSSIAN NOISE WITH THE F1-SCORE AS THE METRIC

Methods	Noise Level				
	0	5	10	15	20
SiamCRNN-50 [30]	82.71	81.54	78.26	67.87	43.08
DSIFN [50]	84.07	83.45	81.85	74.96	56.83
ChangeFormerV6 [13]	80.71	80.32	79.11	75.88	73.26
SwinSUnet [15]	85.60	83.52	78.03	72.19	57.44
MambaBCD [22]	88.39	87.13	84.78	82.50	79.26
HAM-CD (Ours)	**91.54**	**90.88**	**89.95**	**88.74**	**86.55**

TABLE XV

COMPARISON OF ROBUSTNESS AGAINST DIFFERENT LEVELS OF GAUSSIAN BLUR WITH THE F1-SCORE AS THE METRIC

Methods	Standard Deviation				
	0	0.5	1.0	1.5	2.0
SiamCRNN-50 [30]	82.71	80.69	77.45	47.89	24.61
DSIFN [50]	84.07	80.03	78.76	44.76	25.22
ChangeFormerV6 [13]	80.71	79.62	76.39	70.74	59.90
SwinSUnet [15]	85.60	84.54	83.31	78.02	70.08
MambaBCD [22]	88.39	87.35	86.12	84.65	78.72
HAM-CD (Ours)	**91.54**	**91.32**	**90.45**	**88.33**	**84.40**

TABLE XVI

COMPARISON OF ROBUSTNESS AGAINST DIFFERENT INPUT IMAGE RESOLUTIONS WITH THE F1-SCORE AS THE EVALUATION METRIC

Methods	Input Spatial Resolution				
	128^2	192^2	256^2	320^2	384^2
SiamCRNN-50 [30]	68.12	81.75	82.71	81.04	77.95
DSIFN [50]	69.37	83.02	84.07	78.64	68.53
ChangeFormerV6 [13]	62.25	77.83	80.71	77.01	75.82
SwinSUnet [15]	70.88	84.47	85.60	84.33	83.29
MambaBCD [22]	75.12	86.55	88.39	84.65	85.02
HAM-CD (Ours)	**85.06**	**89.58**	**91.54**	**90.38**	**89.57**

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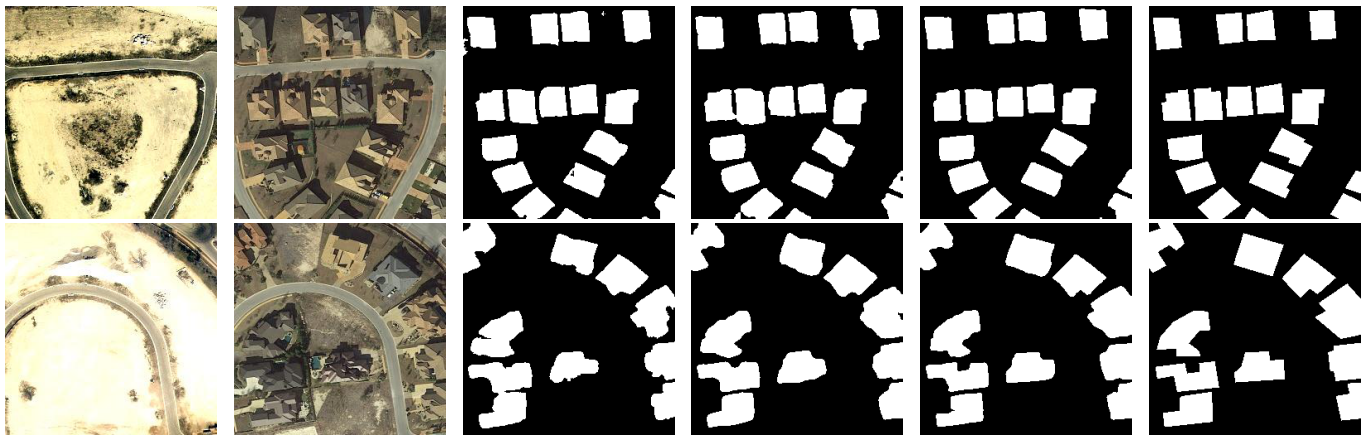


Fig. 6. Visualization of the change maps detected under illumination variations on the LEVER-CD dataset. (a) Pretime. (b) Posttime. (c) ChangeFormer. (d) MambaBCD. (e) Ours. (f) GT.

ChangeFormerV6, SwinSUNet, and MambaBCD demonstrate much stronger stability, with $F1$-score reductions of 25.78%, 18.13%, and 10.94%, respectively. Our proposed HAM-CD model proves to be even more robust, experiencing a minimal decrease of just 7.14%. Third, we compare the robustness of different models trained at a single resolution and test at different resolutions, of which the results are summarized in Table XVI. As shown, when the input resolution increases from 256 $\times$ 256 to 384 $\times$ 384, the $F1$-score of our HAM-CD experiences a minimal reduction of merely 2.15%. Under a more challenging input resolution (128 $\times$ 128), all comparative methods exhibit substantial performance deterioration. Notably, SwinSUNet, ChangeFormerV6, and MambaBCD all exhibited substantial performance declines in $F1$-score, each exceeding 12%. In contrast, our HAM-CD demonstrates significantly stronger robustness, with a comparatively lower reduction of 6.48%. These results across varying input spatial resolution validate the superior resilience and architectural advantage of the proposed HAM-CD framework. The robustness of the HAM-CD method against noise, blur degradations, and varied input resolution is attributed to the design of the HAM module. By enhancing global–local modeling capabilities, this module enables the effective capture of change details and the comprehensive utilization of information from less-degraded image areas under degraded scenarios. These experiments and analyses consistently confirm the robustness of the proposed HAM-CD framework.

G. Effectiveness Under Illumination Variations

In this section, we devote ourselves to validating the robustness of our proposed HAM-CD under varying illumination conditions. We have not explicitly validated the model's adaptability to illumination variations on dedicated datasets. However, as the three employed representative benchmark datasets inherently incorporate scenarios with illumination variations, our experimental evaluations on these benchmarks indirectly demonstrate the model's adaptability to such environmental factors. We demonstrate some qualitative comparison results from the three employed datasets in Fig. 6.

TABLE XVII
EVALUATION OF CROSS-MODAL CD ON THE CALIFORNIA DATASET [61]

Methods	Rec	Prec	OA	F1	IoU
ChangeFormer [13]	81.84	89.66	97.20	85.28	76.83
E-Unet [62]	91.36	89.88	99.45	90.54	82.71
DTCDN [63]	92.93	91.55	99.58	92.17	85.53
GLCD-DA [64]	96.03	95.95	99.66	95.99	92.54
MambaBCD [22]	96.04	96.79	99.70	96.41	93.08
Ours	**96.68**	**97.13**	**99.74**	**96.91**	**94.00**

As shown, the two images captured at different times for the CD input exhibit significant illumination variations. Even under such conditions, our method achieves more accurate detection results compared to mainstream approaches such as ChangeFormer [13] and MambaBCD [22]. This demonstrates the effectiveness and robustness of our HAM-CD method in handling scenes with pronounced illumination changes.

H. Evaluation Under Water-Body Interference

We qualitatively analyzed the performance of different methods in handling water-body interference and demonstrated the experimental results in Fig. 7. The experimental results demonstrate that the proposed method can successfully segment change regions that more closely align with the GT, even in the presence of extensive water bodies and their color variations in the background. Although these results demonstrate a degree of efficacy in mitigating water-body interference, there remains considerable potential for further enhancement to achieve more precise CD.

I. Robustness on Multimodal CD

To further assess the generalization ability and robustness of HAM-CD with heterogeneous cross-modal data in complex real-world scenarios, we conduct additional experiments on the California dataset [61]. This dataset is particularly challenging as it captures a large-scale flood event in California, USA,

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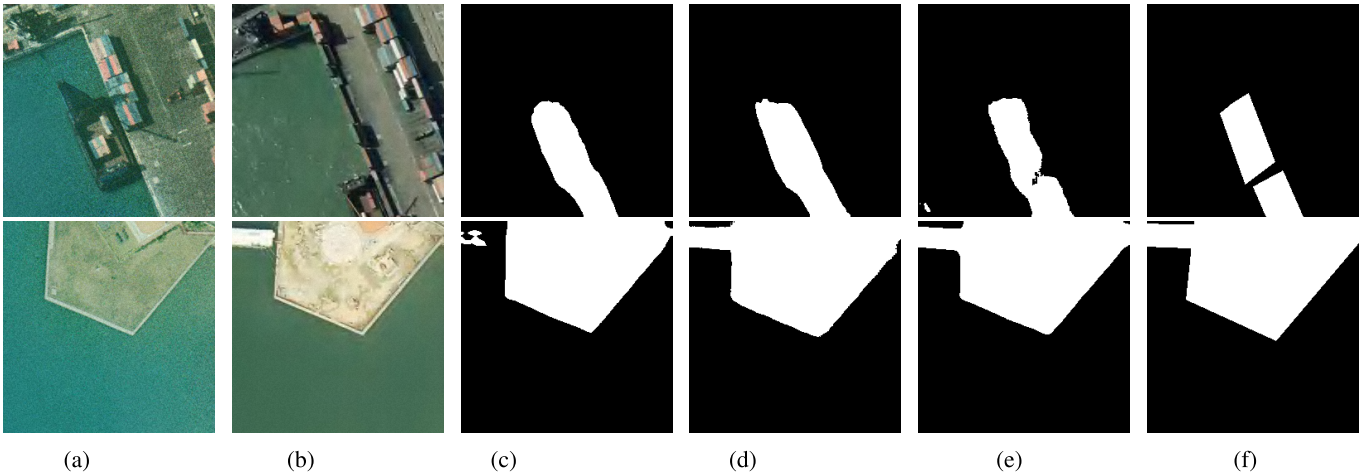


Fig. 7. Visualization of the change maps detected under water bodies interference on the SYSU-CD dataset. (a) Pretime. (b) Posttime. (c) ChangeFormer. (d) MambaBCD. (e) Ours. (f) GT.

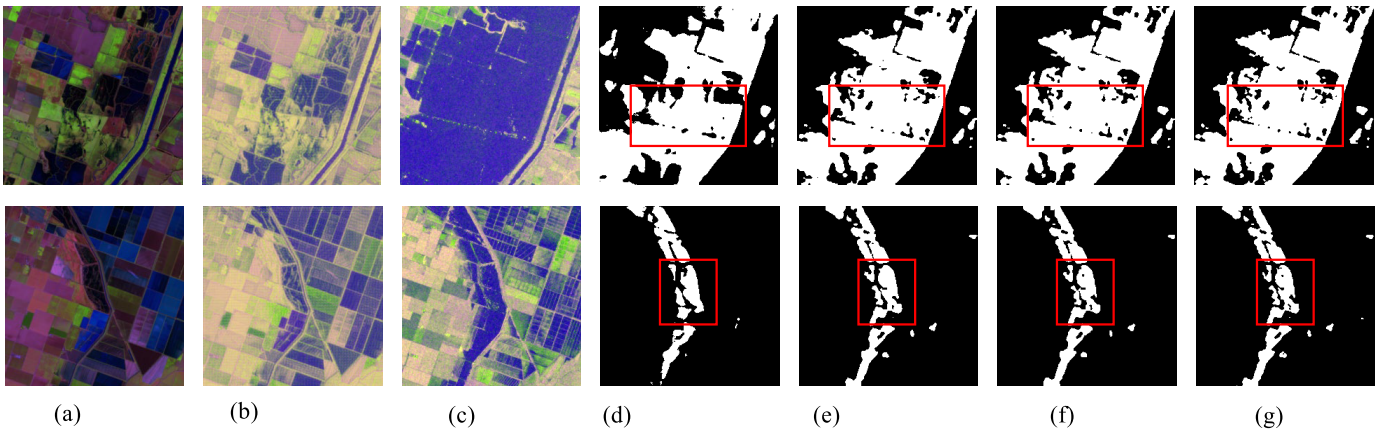


Fig. 8. Visual comparison results of different CD methods on the cross-modal California dataset [61]. (a) Optical. (b) SAR-like. (c) SAR. (d) ChangeFormer. (e) MambaBCD. (f) Ours. (g) GT.

characterized by severe terrain variations and significant spectral discrepancies between preevent and postevent imagery. Unlike typical homogeneous datasets, the California dataset consists of heterogeneous cross-modal data, specifically a Landsat-8 multispectral image acquired in January 2017 and a Sentinel-1A SAR image acquired in February 2017. The SAR image comprises three channels, specifically the VV and VH polarizations along with their intensity ratio, providing a unique modality for CD evaluation. he spatial resolution of the two images is 2000 $\times$ 3500. The GT data were derived from two other single-polarization Sentinel-1 images captured at the same time. For dataset preparation, we follow DTCDN [63] and GLCD-DA [64], where we crop the original images into pairs of 256 $\times$ 256 with a 50% overlap rate. Then, the dataset is divided into training and testing sets, where the eight neighboring images around each image in the testing set are excluded from the training set. We take 80% of the data for training, and the rest 20% for testing. For the training and evaluation, we adopt the same two-stage strategy used in DTCDN [63] and GLCD-DA [64]. In the first stage, we employ the CycleGAN [65] as the image transfer network to transfer the optical image from the optical domain to the SAR domain. As discussed in GLCD-DA [64], the CD performance is superior when converting optical imagery to SAR. The produced SAR images, along with the original SAR remote sensing images, are subsequently fed into the CD network. Then, in the second stage, we employ our HAM-CD as the CD network to produce the binary CD map. We compare our HAM-CD method with several outstanding optical-SAR CD methods, including ChangeFormer [13], E-Unet [62], DTCDN [63], GLCD-DA [64], and MambaBCD [22]. To ensure fair comparison, all methods employ the same batch size and number of training epochs as our method, while the remaining parameters follow the original configurations specified in their respective articles. We summarize the CD results on the California dataset in Table XVII. As shown, ChangeFormer [13] exhibits the lowest $F1$-score, which may be due to the significant distributional differences between the heterogeneous images. E-Unet [62], DTCDN [63], and GLCD-DA [64] are specifically designed for heterogeneous image CD. Among them, GLCD-DA [64] achieves better performance, attaining an $F1$-score of 95.99% and an IoU of 92.54%. MambaBCD [22] demonstrates the second-best results, with an $F1$-score and IoU of 96.41% and 93.08%,

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Fig. 9. Failure cases of the change maps generated by our HAM-CD on the SYSU-CD dataset. (a) Pretime. (b) Posttime. (c) Ours. (d) GT.

respectively. The evaluation confirms the effectiveness of the MambaBCD for heterogeneous CD. However, our HAM-CD outperforms MambaBCD, with an improvement of 0.92% and 0.50% in IoU and $F1$-score. This improvement can be attributed to the parallel integration of TAB and TAM. Specifically, this design enables more effective global semantic alignment and local feature preservation between optical and SAR domains, which is crucial for accurate CD in cross-modal scenarios. The visual results on the California dataset are presented in Fig. 8. As shown, all compared methods exhibit a certain number of false and missed detections. Compared to other methods, our HAM-CD achieves the best structural preservation, maintaining higher shape integrity and producing results with clearer flooded area boundaries, as illustrated in the region marked by the red rectangle. These findings empirically confirm that HAM-CD possesses strong generalization ability for heterogeneous cross-modal CD.

V. Limitations and Future Works

The proposed HAM-CD achieves SOTA performance across all three benchmark datasets compared to existing approaches. However, it struggles to accurately delineate change boundaries under certain challenging, complex scenarios, such as dense vegetation and shadow areas. We visualize two characteristic failure cases in Fig. 9. We attribute these limitations to the severe suppression of salient change signatures by complex background textures and adverse illumination conditions. In regions with dense vegetation, complex spatial textures act as high-frequency background noise that can suppress the saliency of actual change features. Similarly, in shadow-occluded areas, the low contrast and spectral degradation lead to a significant attenuation of salient features. This degradation makes it difficult for the model to distinguish between the complex background and true semantic changes, often resulting in false negatives or boundary blurring. These issues are further aggravated by the scarcity of scene-specific datasets that cover such complex scenarios. To address these limitations, our future work will focus on developing adaptive feature decoupling modules designed to isolate environmental noise from semantic change signals, thereby mitigating the feature suppression effect. We also intend to incorporate a brightness-aware module based on contrastive learning to extract representations that are invariant to light intensity and enhance feature discrimination. By doing so, we aim to enhance the model's robustness against brightness interference in global features, thereby maintaining superior performance under varying seasonal, illumination, or shadow conditions. Furthermore, we plan to construct specialized datasets using generative adversarial networks to synthesize diverse vegetation and shadow conditions, ensuring the model encounters sufficient hard scenarios during training. Regarding practical deployment, although HAM-CD already benefits from the inherent efficiency of the TAB, which achieves linear complexity by computing cross-covariance across channels, there remains significant potential for further optimization. Specifically, we intend to explore structural pruning targeted at the redundant channel-wise correlations within the TAB to further reduce computational overhead without compromising the model's ability to capture global dependencies. Besides, we plan to employ knowledge distillation to transfer high-level semantic features to lightweight student models. These efforts will effectively bridge the gap between high-performance architecture and edge-device requirements, facilitating the seamless deployment of HAM-CD in real-world, large-scale remote sensing monitoring systems.

VI. Conclusion

This study presents HAM-CD, a novel hybrid architecture integrating Mamba and SA for remote sensing binary CD. Building upon spatial-Mamba's efficient global and local context modeling capability, we introduce a HAM decoder to effectively capture spatial–temporal relationships. The HAM block seamlessly fuses the TAM module to strengthen local pixel dependencies and the TAB to enhance global dependencies. To further optimize feature representation, the proposed ICSF block enhances feature integration across channel and spatial dimensions through joint modeling of their semantic differences. Evaluation results on three benchmark datasets show that HAM-CD outperforms both pure Mamba- and Transformer-based methods, proving its efficacy for multitemporal remote sensing data analysis. Besides, the evaluation results on the heterogeneous cross-modal California dataset validate our model's versatility and generalization ability across real-world remote sensing scenarios. The successful integration of attention mechanisms with Mamba architecture offers a new direction for modeling complex spatial–temporal patterns in CD tasks.

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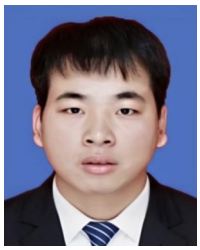
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